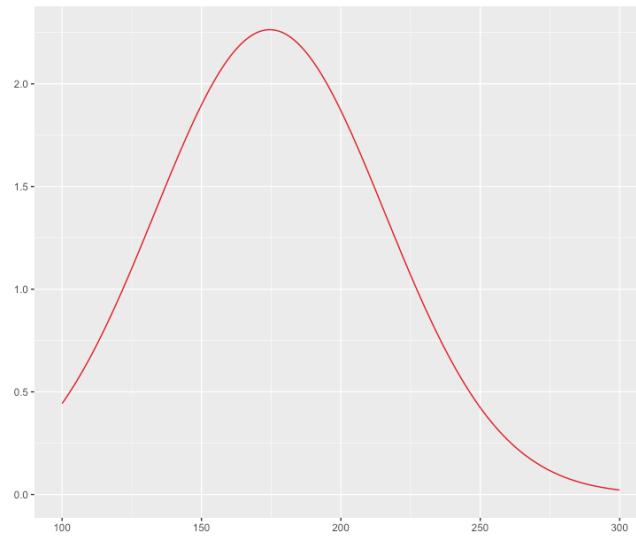
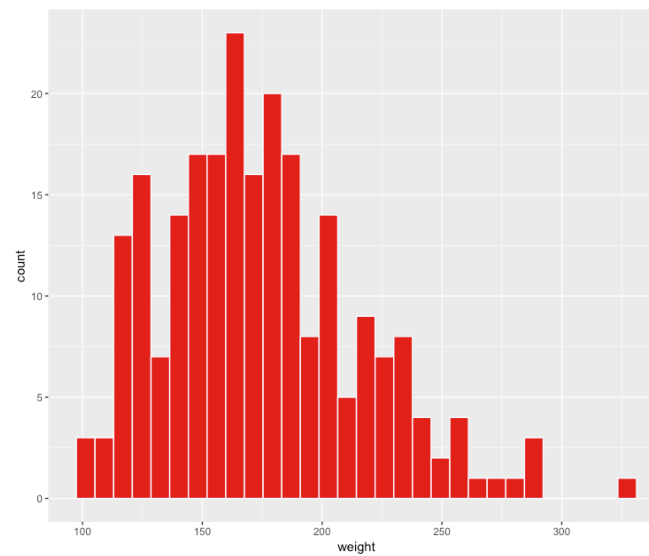


6. Statistical inference

Statistical inference



How can we estimate the
parameters of the population
(expectation, variance, ..) ...



... using the sample distribution?

Random variables and samples

- Parameters associated to the RV are indicated in greek letters
- Parameters associated to sample are in roman letter

Random variable	Sample
random variable X	realizations $x_1, x_2, \dots, x_n$
probability distribution $p(k)$ density distribution $f(X)$	histogram
Expectation $E(X) = \mu$	Mean / median $\bar{x} = m$
Variance $Var(X) = \sigma^2$	Sample variance $Var(x_i) = s^2$

Can we use m and s to estimate μ and σ ?

Infering the expectation

- Example: **number of children per family**
→ we conduct a survey and interview every day n families
- Random variable : **$N_i = \{\text{number of children in the } i\text{-th family}\}$**
- The N_i are random variables, independent of each other, and identically distributed ("i.i.d. = independent, identically distributed")

$$M_n = \frac{1}{n} \sum_{i=1}^n N_i$$

- M_n is a **new random variable**
→ estimate its probability distribution $p(k)$, expectation and variance
- $M_n =$ random variable
realization = **m_n^k** (k = day of the measurement)

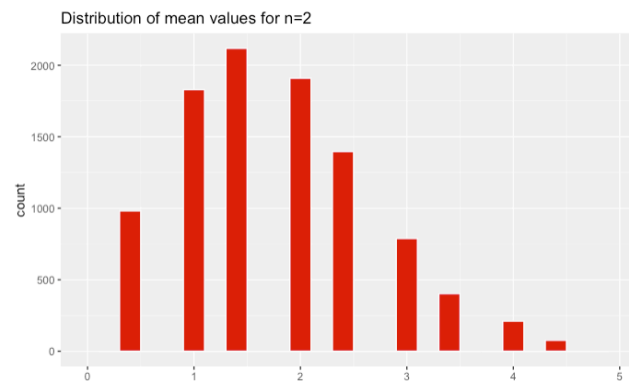
Example n=2

	Day_1	Day_2	Day_3	Day_4	Day_5	Day_6	Day_7	Day_8	Day_9	Day_10
Family_1	1	0	4	4	3	2	4	2	0	0
Family_2	3	2	1	1	2	2	4	3	1	1

...

	Day_1	Day_2	Day_3	Day_4	Day_5	Day_6	Day_7	Day_8	Day_9	Day_10
m ₂	2	1	2,5	2,5	2,5	2	4	2,5	0,5	0,5

...



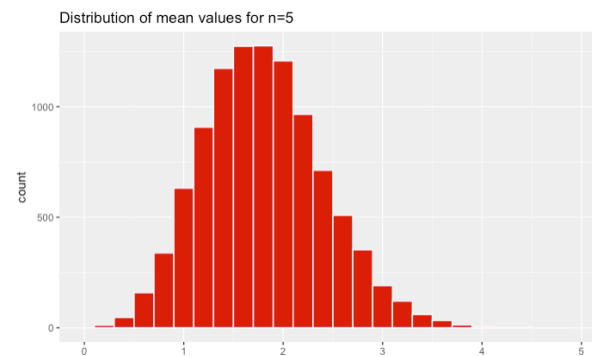
Example n=5

	Day_1	Day_2	Day_3	Day_4	Day_5	Day_6	Day_7	Day_8	Day_9	Day_10
Family_1	1	0	4	4	3	2	4	2	0	0
Family_2	3	2	1	1	2	2	4	3	1	1
Family_3	1	4	2	0	2	2	2	1	1	2
Family_4	3	2	2	1	6	1	3	1	1	1
Family_5	4	1	0	4	2	0	0	1	0	3

...

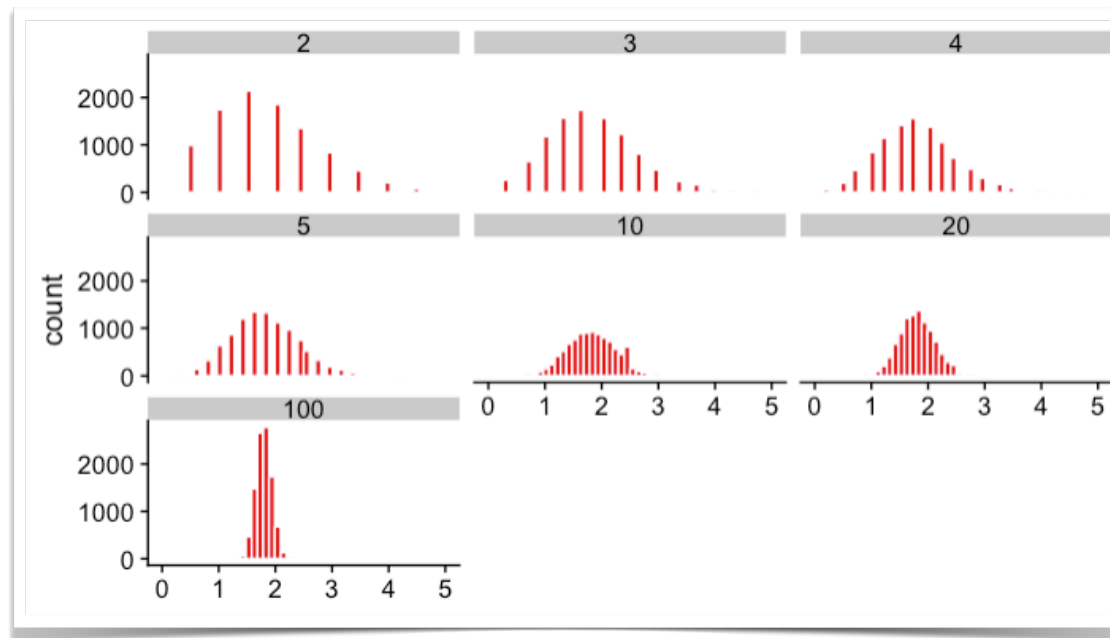
	Day_1	Day_2	Day_3	Day_4	Day_5	Day_6	Day_7	Day_8	Day_9	Day_10
m_5	2,4	1,8	1,8	2	3	1,4	2,6	1,6	0,6	1,4

...



Distribution of mean values

- For increasing sample size n :
 - distribution of mean values **tends to a normal distribution**
 - distribution of mean values **becomes narrower**
- **less fluctuations of the mean between samples**



Distribution of means

- For each value of n , compute
 - mean of the distribution (red dots)

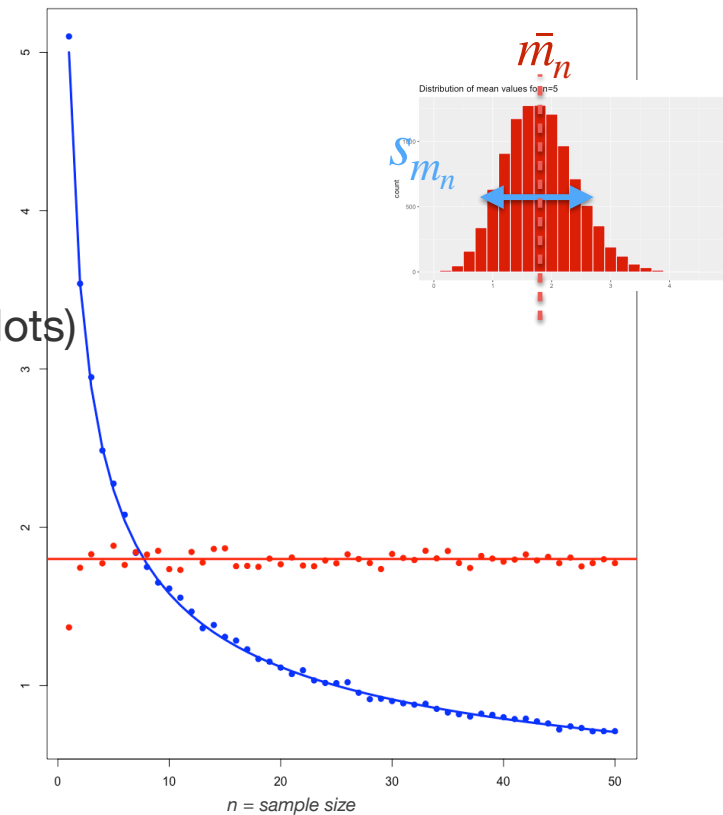
$$\bar{m}_n \sim \mu$$

- standard deviation of the distribution (blue dots)

$$s_{m_n} \sim \sigma/\sqrt{n}$$

- this is the **standard error**

*mean of the sample is a good estimator
of the expectation*



$$M_n \longrightarrow \mathcal{N}(\mu, \sigma/\sqrt{n})$$

Central Limit Theorem

or : why is the SND the Queen of distributions...

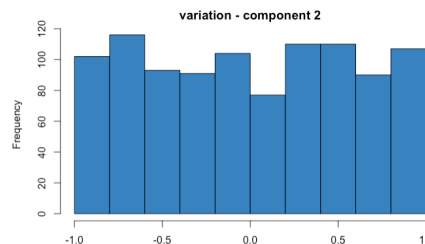
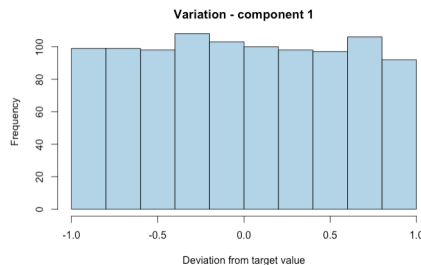
- $\{X_1, X_2, \dots, X_n\}$ n random variables such that
 - independent
 - identically distributed
 - $E[X_i] = \mu$; $Var(X_i) = \sigma^2$
- then the following holds:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{n \rightarrow \infty} \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$$

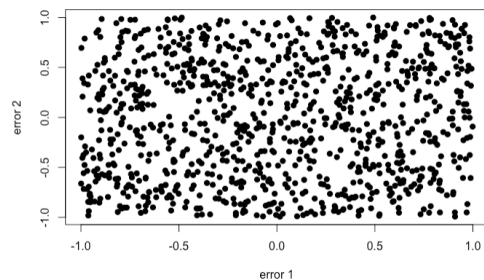
$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \xrightarrow{n \rightarrow \infty} \mathcal{N}(0,1)$$

Central Limit Theorem

- **Example:** effect of multiple sources of errors
- Production process - multiple sources of error X_i on the final weight of the product
- Assume these errors are i.i.d (independent, identically distributed - for example uniformly distributed)



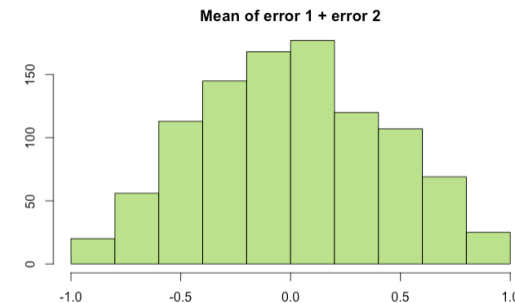
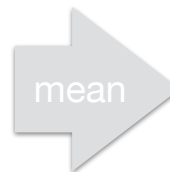
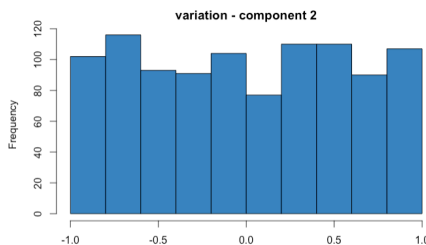
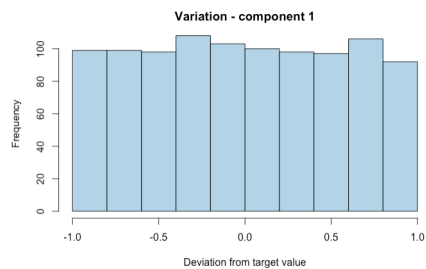
identically distributed
(uniform)



independent (no correlation)

Central Limit Theorem

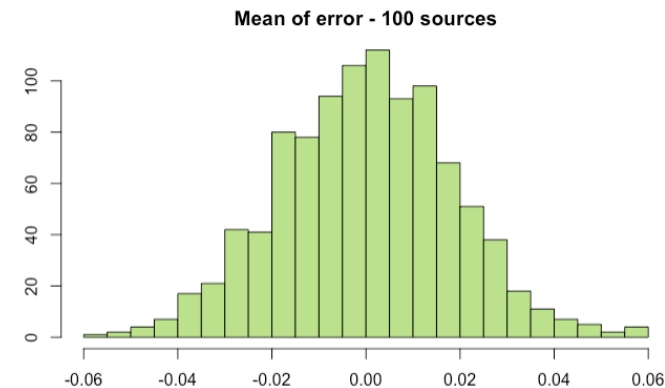
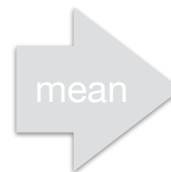
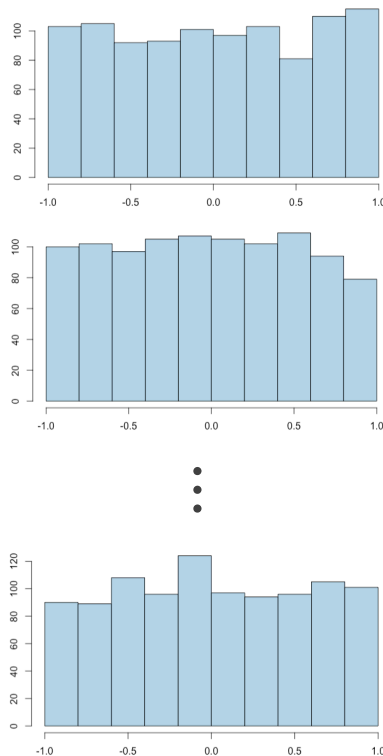
- **Example:** effect of multiple sources of errors
- Total error on final product = mean of errors (for example on weights)



*This is no longer
a uniform distribution!*

Central Limit Theorem

- **Example:** effect of multiple sources of errors
- Total error on final product = sum of errors (for example on weights)



***Converges to a normal
distribution!***

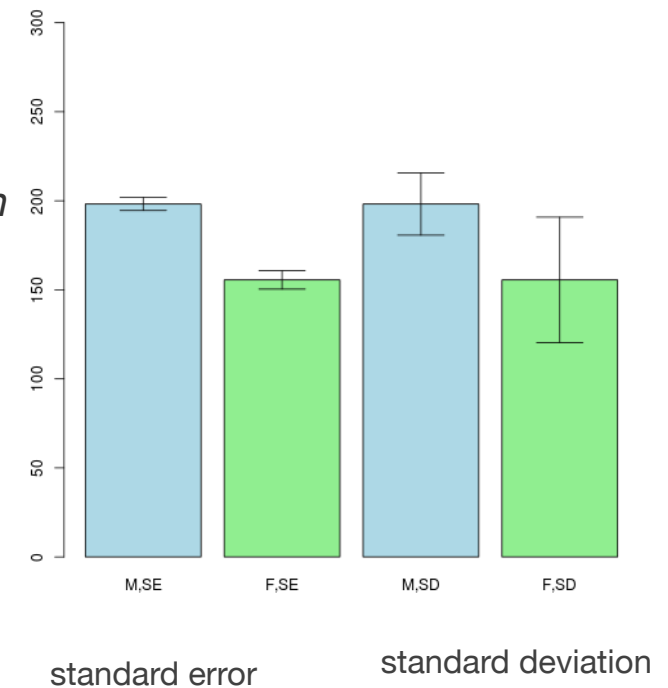
Standard error vs. Standard deviation

- **Standard deviation** describes the *spread in a distribution*

Standard deviation : σ

- **Standard error** measures the *spread in the estimation of the mean*

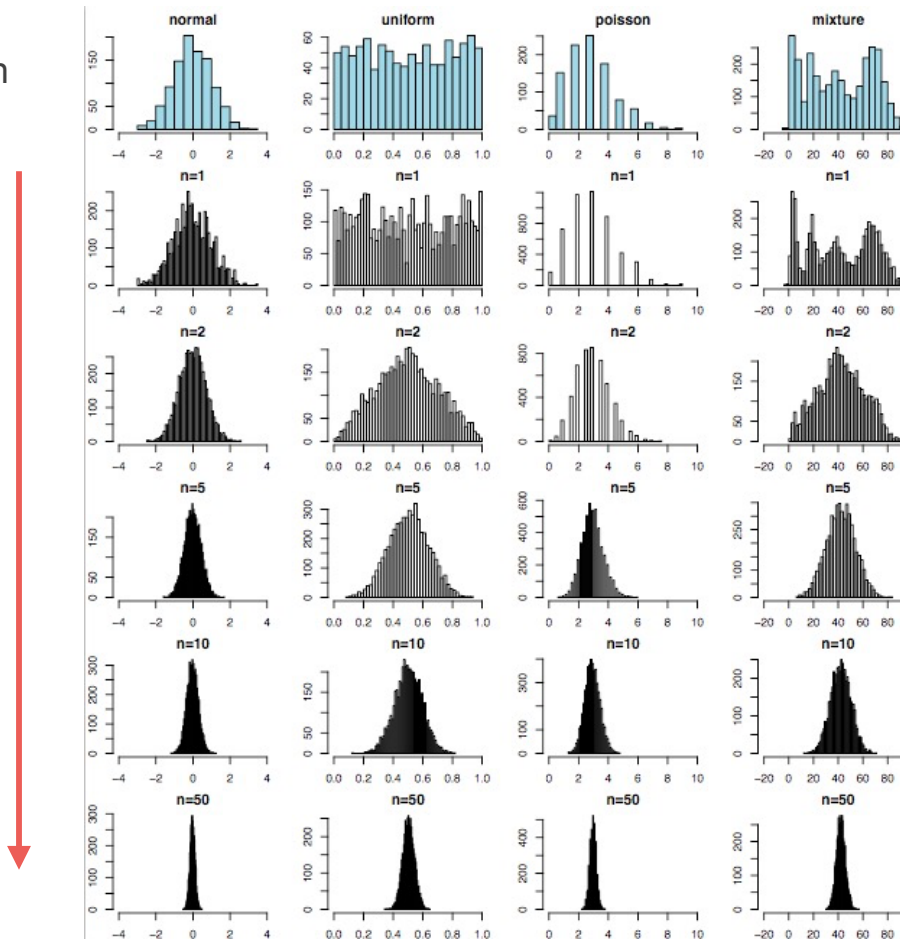
$$\text{Standard error} : \frac{\sigma}{\sqrt{n}}$$



Estimation of expectation value

initial distribution

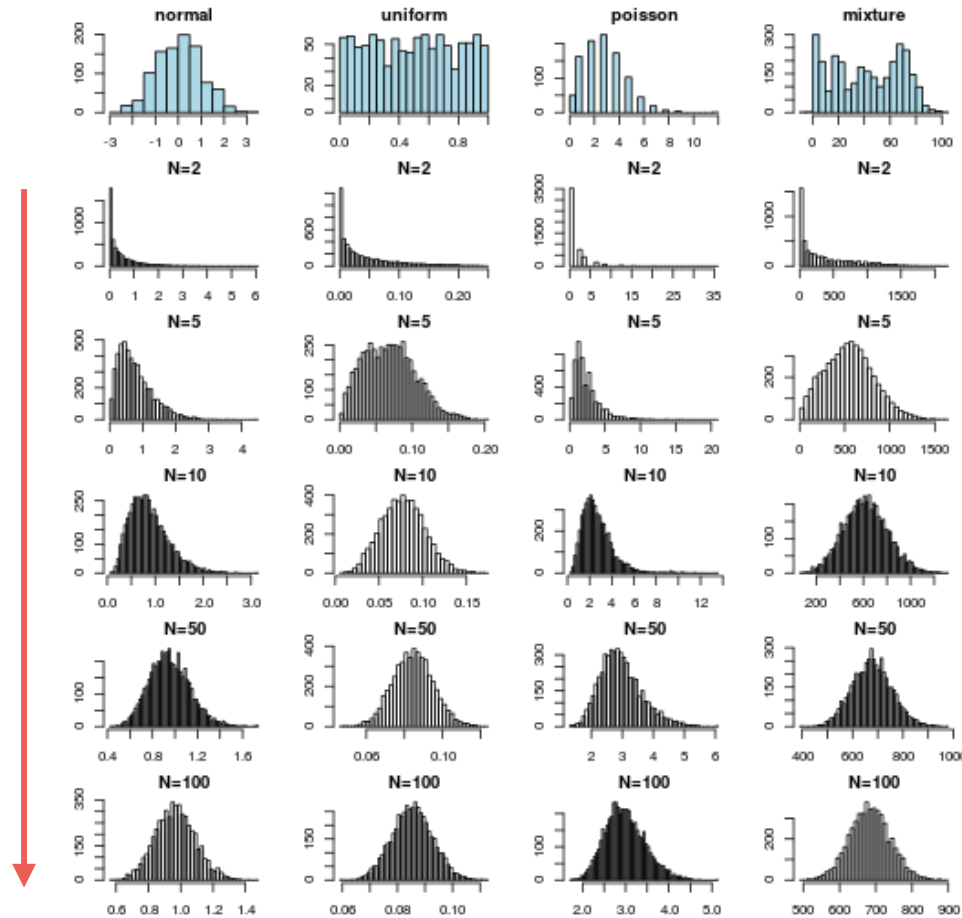
increasing
sample size



Estimation of variance

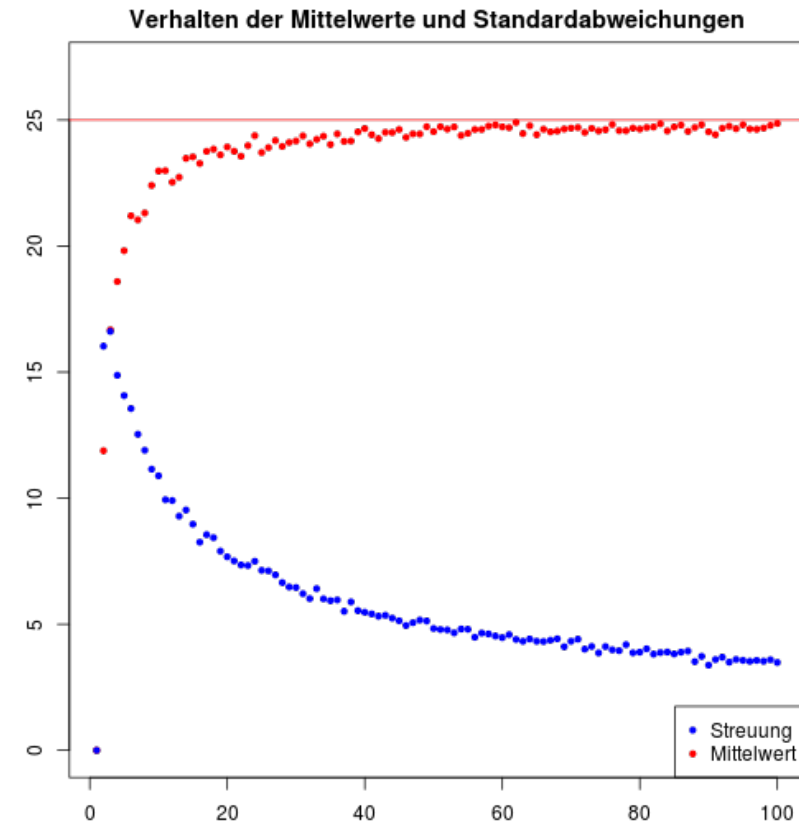
$$Var_n = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

increasing
sample size



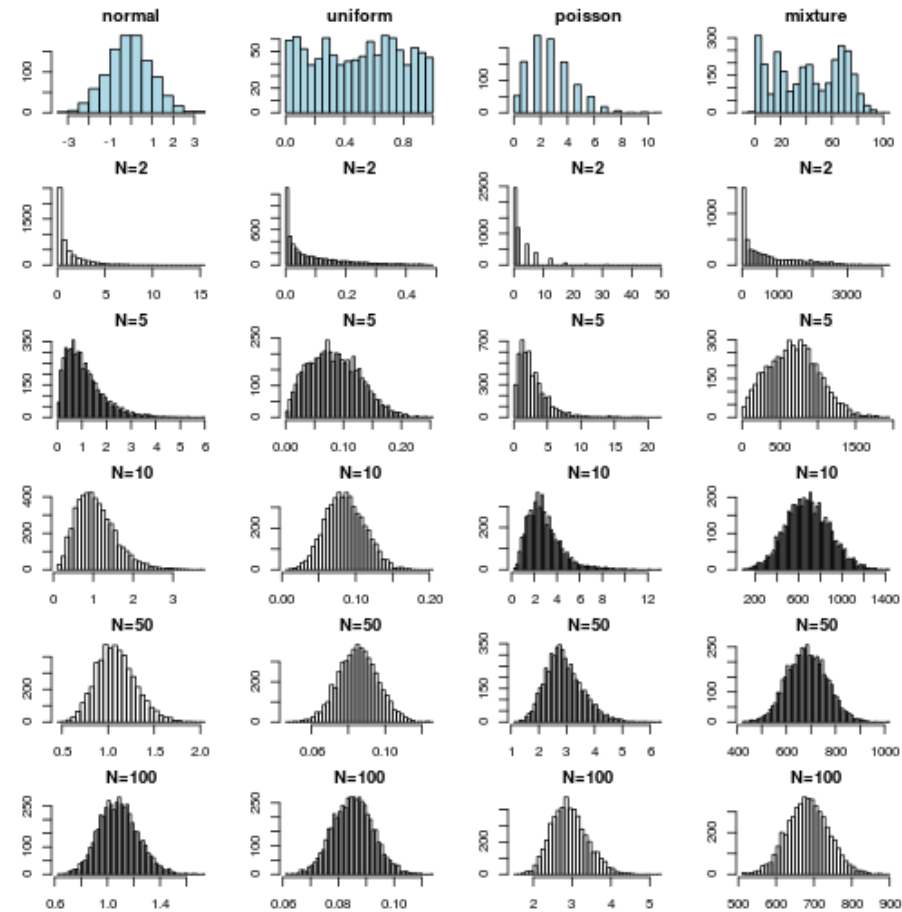
Estimation of variance

- Estimated sample variance
- converges **only asymptotically** to the true variance of the random variable



Estimation of variance

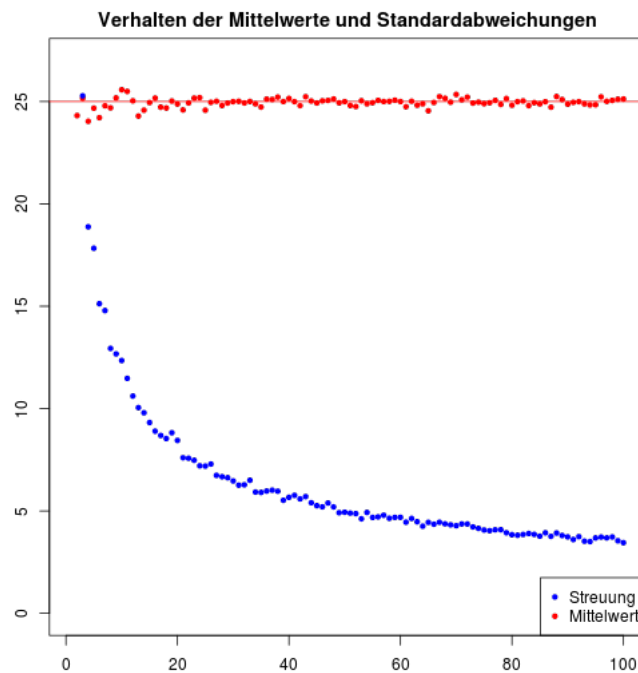
$$Var_n = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$



Estimating variance

$$Var_n = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 \longrightarrow Var_n = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

is an **unbiased estimator** of the population variance



$$Var_n = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

Confidence interval

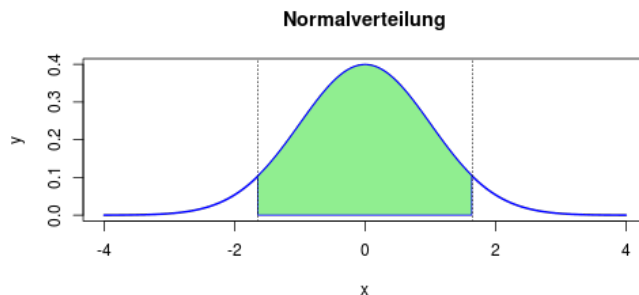
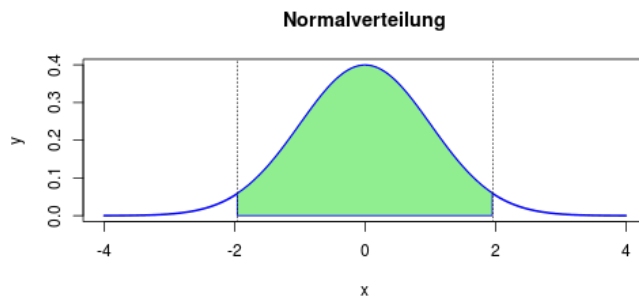
- large sample:

$$M_n \xrightarrow{n \gg 1} \mathcal{N}\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

Central Limit Theorem

$$M_n \sim \mathcal{N}\left(\mu, \frac{\sigma}{\sqrt{n}}\right); \frac{M_n - \mu}{\sigma/\sqrt{n}} \sim \mathcal{N}(0, 1)$$

realization of the
random variable!



- in 95% of the cases, $\left| \frac{m_n - \mu}{\sigma/\sqrt{n}} \right|$ is smaller than $t_{95}=1.96$
- 95% confidence interval:

$$\mu \in \left[m_n - t_{95} \frac{\sigma}{\sqrt{n}} ; m_n + t_{95} \frac{\sigma}{\sqrt{n}} \right]$$

- Problem: we do not know the value of the population standard deviation σ
- Replace by its estimator

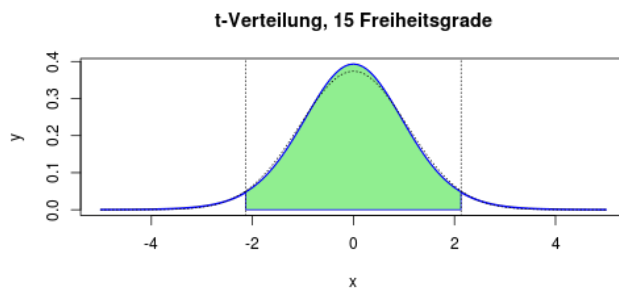
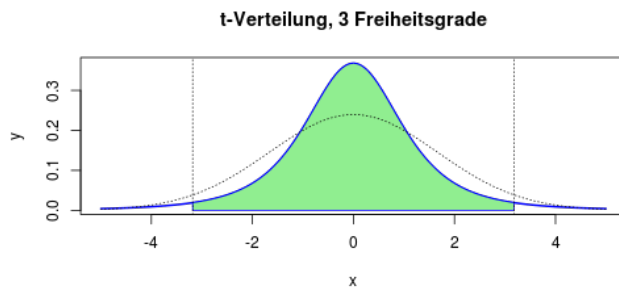
$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$

Confidence interval

- small sample:

$$M_n \xrightarrow{n \gg 1} \mathcal{N}\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

$$\frac{M_n - \mu}{\sigma/\sqrt{n}} \sim t_{n-1}$$



- in 95% of the cases, $\left| \frac{m_n - \mu}{\sigma/\sqrt{n}} \right|$ is smaller than $t_{95,n-1}$
- 95% confidence interval:

$$\mu \in \left[m_n - t_{95,n-1} \frac{\sigma}{\sqrt{n}} ; m_n + t_{95,n-1} \frac{\sigma}{\sqrt{n}} \right]$$
- Problem: we do not know the value of the population standard deviation σ
- Replace by its estimator

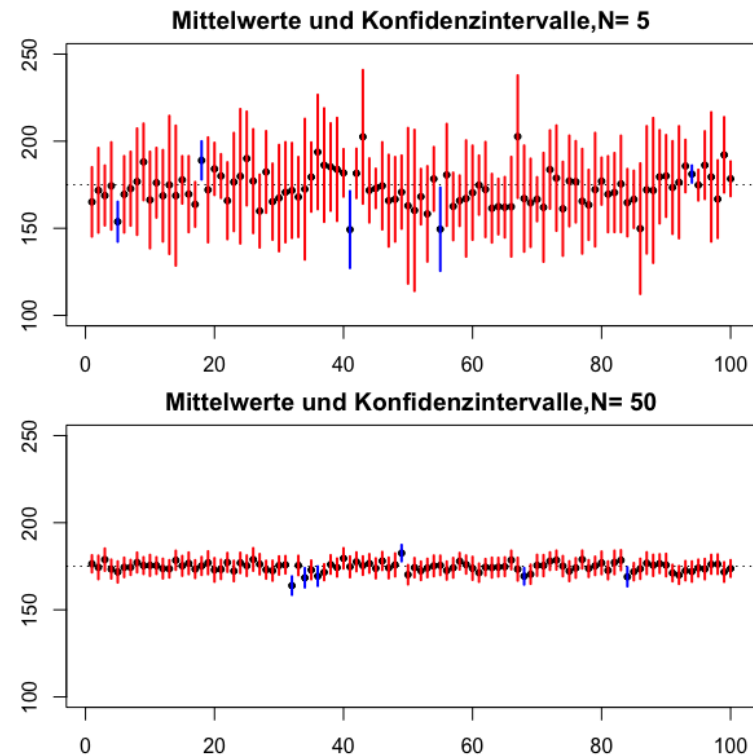
$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$

Critical values for t-distribution

df	t ₉₅	t ₉₀
2	4.302653	2.919986
3	3.182446	2.353363
4	2.776445	2.131847
5	2.570582	2.015048
8	2.306004	1.859548
10	2.228139	1.812461
15	2.131450	1.753050
20	2.085963	1.724718
50	2.008559	1.675905
100	1.983972	1.660234
200	1.971896	1.652508
Normal	1.959964	1.644854

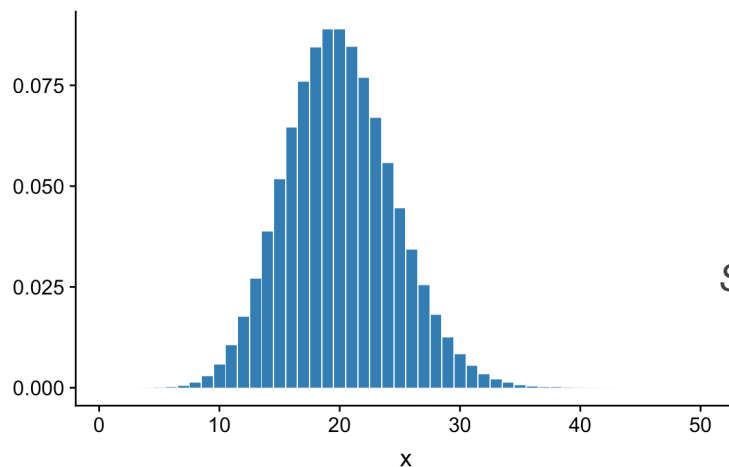
Interpretation of confidence interval

- 95% confidence interval
 - 100 samples with n elements
 - sample distribution with expectation value μ
 - estimate mean and CI
- True expectation value lies in 95% of the cases within the confidence interval



Confidence interval for Poisson distributions

- I find $k=20$ raisins in my raisin bun
→ what's in the recipe?
- Number of raisins N per bun is a **random variable**, described by a Poisson distribution



$$N \sim \text{Pois}(\lambda = k)$$

$$\text{Pois}(k) \xrightarrow{k \gg 1} \mathcal{N}(k, \sqrt{k})$$

Sample size $n=1$:

$$\mu \in \left[m_n - t_{95} \frac{\sigma}{\sqrt{n}} ; m_n + t_{95} \frac{\sigma}{\sqrt{n}} \right]$$

$$\mu \in [k - t_{95} \sqrt{k} ; k + t_{95} \sqrt{k}]$$

$$11.2 \leq \mu \leq 28.8$$

Confidence interval Poisson distribution

How to reduce the confidence interval?

- **eat more buns!** $n=10$ samples with 1 bun
- Compute the mean over the sample

28 17 22 19 18 ...

$$\bar{k} = 20.2$$



$$\mu \in \left[m_n - t_{95} \frac{\sigma}{\sqrt{n}} ; m_n + t_{95} \frac{\sigma}{\sqrt{n}} \right]$$

$$\mu \in \left[\bar{k} - t_{95} \frac{\sqrt{\bar{k}}}{\sqrt{10}} ; \bar{k} + t_{95} \frac{\sqrt{\bar{k}}}{\sqrt{10}} \right]$$

$$17.4 \leq \mu \leq 22.9$$

Week 5 - stop and think

*First think about the questions on a sheet of paper;
then use your favorite AI to generate a small R code to illustrate these questions!*

- **Question 1:** According to the Central Limit Theorem, if we have n independent and identically distributed random variables $X_1, X_2, \dots, X_n$ with mean μ and variance σ^2 , the distribution of the sample mean $\bar{X}$ converges to:
 1. A uniform distribution with mean μ
 2. A Poisson distribution with parameter $\lambda = \mu$
 3. A normal distribution with mean μ and variance σ^2/n
 4. A t-distribution with $n-1$ degrees of freedom
- **Question 2:** Explain the difference between standard deviation and standard error. Why does the standard error decrease as sample size increases, and what does this tell us about our estimate of the population mean?
- **Question 3:** When calculating a 95% confidence interval for small samples, we use the t-distribution instead of the normal distribution. Looking at the critical values table in the lecture, explain what happens to the t-values as the sample size increases and why this makes sense.
- **Question 4:** Increasing the sample size will make the confidence interval smaller or larger? The 90% CI is larger or smaller than the 95% CI?